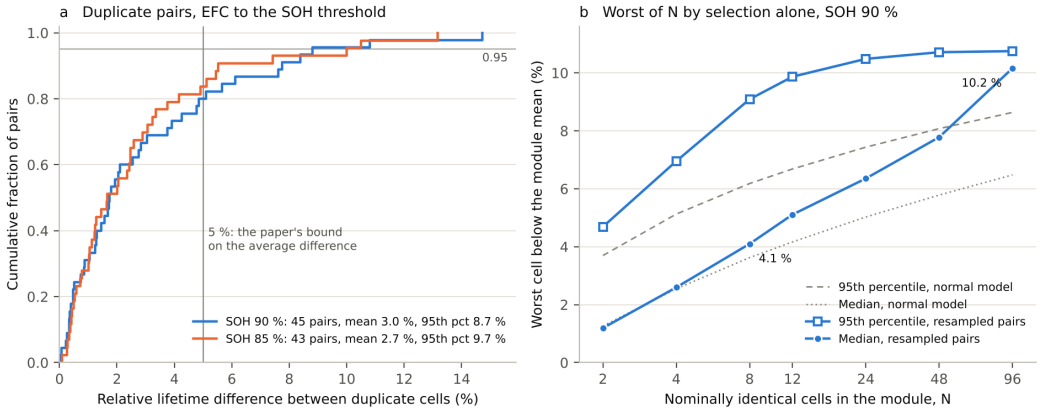
The cell-to-cell lifetime floor in the Geslin et al. dynamic-cycling dataset

Siddharth Satte. A pre-registered re-analysis of the 45 duplicate pairs in Geslin, Xu, Ganapathi, Moy, Chueh and Onori, Nature Energy 10, 172-180 (2025), from the data and code the authors released (dataset doi.org/10.25740/td676xr4322, CC BY 4.0).

8.7 %: the 95th percentile of the lifetime difference between duplicate cells at 90 % SOH (percentile bootstrap over pairs, 5.7 to 13.9 %).

4.1 %: the median shortfall of the worst of eight nominally identical cells below the module mean lifetime, by selection alone, at 90 % SOH (2.8 to 5.6 %).



(a) Cumulative distribution of the relative difference in EFC to 90 % and 85 % SOH between the two cells of each duplicate pair; the vertical line marks the 5 % bound the paper states for the average. (b) Shortfall of the worst of N identical cells below the module mean at 90 % SOH by selection alone, resampling the 90 measured pair deviations (blue) or drawing from a normal model with the fitted sigma (gray).

Result

The average pair difference at 90 % SOH is 3.0 % (median 1.7 %), consistent with the paper's statement that it is below 5 %; recomputing the paper's Fig. 2c group averages with its released code gives all nine below 5 % (largest 4.9 %). The floor sits in the tail: 9 of 45 pairs differ by more than 5 % and 2 by more than 10 % (largest 14.7 %, periodic profile E at C/5), and the 95th percentile is 2.9 times the mean, a ratio that 45 pairs of normal noise reach in 2.6 % of draws, so the resampled worst-of-8 shortfall (4.1 %) exceeds the normal model's (3.6 %). The worse cell of a pair changes with the lifetime or fade-rate estimator in 36 of 45 pairs, pairs hold 3.0 % of the lifetime variance against 97.0 % between protocols, and removing the protocols the authors excluded moves both headline numbers by less than 0.3 points.

Method

1. Data: the processed diagnostics released with the paper, 92 commercial SiOx-graphite/NCA cells under 47 protocols at 35 °C; 45 protocols have two cells, and cell_017 and cell_084, whose duplicates failed, are excluded by construction.
2. Lifetime: the EFC at which the C/2 capacity reaches 95, 92.5, 90, 87.5 or 85 % of its first value, after dropping every point that a later diagnostic exceeds (the paper's stated rule), by linear interpolation.
3. Pair difference: |EFC_a - EFC_b| divided by the pair mean at the same threshold; the per-cell noise sigma is the mean difference times sqrt(pi)/2, which is 2.65 % at 90 % SOH.
4. Worst of N: simulated modules of N cells whose deviations are resampled from the 90 pair deviations (half the difference, either sign, times sqrt 2) or drawn from a normal with that sigma; the shortfall is (mean minus minimum) over mean.
5. Checks: twelve lifetime and fade-rate estimators per cell; the variance split within pairs and between protocols; the paper's Fig. 2c recomputed with its own code; a second, independent implementation agrees to the last digit.
6. Pre-registration: seven predictions with bands were committed (409eba9) and pushed before the analysis ran; code, input hashes and results are at github.com/siddharth161204-wq/geslin-floor-pilot, archived as doi.org/10.5281/zenodo.23205413.

Limits

1. One temperature, one cell type, lifetime from capacity only, and 45 pairs: the 95th percentile rests on the three largest pairs, and the worst of 96 (10.2 %) is set by the single largest pair, so it is an extrapolation.
2. The floor includes diagnostic measurement noise and interpolation between diagnostics up to 100 cycles apart, not only intrinsic cell variation; a pair difference is symmetric by construction, so skew in single-cell lifetimes cannot be seen.
3. The lifetime definition matters: the authors' released file of lifetimes at 90 % SOH places four cells at the first threshold crossing, up to 95 EFC earlier, and with that file the 95th percentile at 90 % SOH is 10.8 % (found while reconciling, not pre-registered).

Prediction scorecard

Prediction	Predicted (band)	Measured	Verdict
Mean pair difference, 90 % SOH	3.5 % (2.5 to 5.0)	2.99 %	Held
95th percentile, 90 % SOH	9.0 % (6.5 to 12.5)	8.72 %	Held
95th percentile, 85 % SOH	9.5 % (6.5 to 13.5)	9.74 %	Held
Per-cell sigma, 90 % SOH	3.1 % (2.2 to 4.4)	2.65 %	Held
Worst of 8, median, resampled	4.2 % (3.0 to 6.0)	4.10 %	Held
Worst of 96, median, resampled	8.5 % (6.0 to 12.5)	10.16 %	Held
Worse cell depends on the estimator	70 % (50 to 90)	80 %	Held
Within-pair variance share, 90 % SOH	1.5 % (0.4 to 4.0)	2.99 %	Held
Tail: 95th percentile / mean above 2.33	2.6	2.91	Held

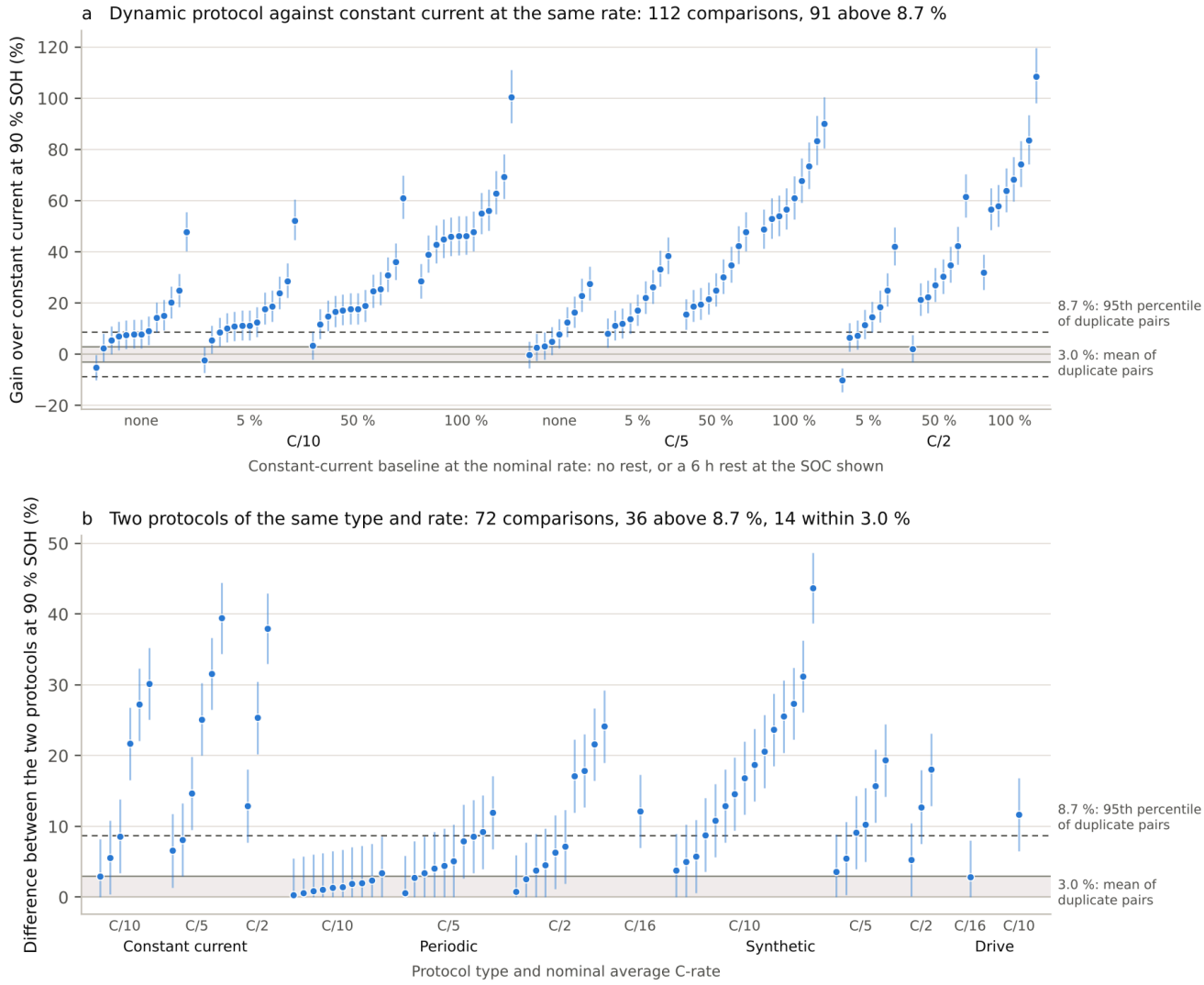
The tail was heavier than predicted: the ratio also passed 2.80, the value only 5 % of normal datasets reach, which I had not expected (unscored); the within-pair share was twice the point prediction.

What it means for the thesis

Under one protocol at 35 °C, the worst of eight nominally identical cells already falls 4.1 % below the module mean lifetime by chance alone, and one duplicate pair in five differs by more than 5 %, so a thermal account of cell-to-cell divergence in a module has to produce gaps beyond this floor before it can be told apart from selection. Because the worse cell of a pair changes with the lifetime estimator in four pairs out of five, the thesis comparison of resolved and lumped module models will fix its lifetime definition in advance and test the worst-cell gap against this floor, not against an average.

Code: MIT. Note, figure and results: CC BY 4.0. Data: Geslin, A., Xu, L., Ganapathi, D., Moy, K., Chueh, W. & Onori, S. Dataset - Dynamic cycling enhances battery lifetime. Stanford Digital Repository (2024), CC BY 4.0.

Addendum (v1.1)



1. Pre-registered (predictions committed in 81cab28 before any comparison was computed): the floor applied at 90 % SOH to the paper's own comparisons, each protocol's lifetime the mean of its two cells with a standard error of sigma divided by sqrt 2, in two families: (a) each dynamic protocol against each constant-current protocol at its nominal rate, 112 comparisons; (b) each pair of protocols of one type and rate, 72.
2. Against all four constant-current baselines 91 of 112 gains exceed 8.7 %, but the share follows the baseline: 30 of 30 against the constant current with a 6 h rest at 100 % SOC (the rests lower those baselines' measured average rate to 0.06 to 0.11 per hour), and 10 of 22 against the constant current without rest, the paper's baseline, where the median gain is 7.9 % and 3 gains lie within plus or minus 3.0 %.
3. The largest gain against the paper's baseline is 47.8 % (40.3 to 55.7 %), Synthetic 1b at C/10, which the paper sets aside because its measured rate is 0.12 per hour; in the paper's own set 8 of 20 gains exceed 8.7 % and the largest is 27.5 % (21.1 to 34.3 %), Synthetic 1a at C/5.
4. Of the 72 comparisons within a type and rate, 36 exceed 8.7 % and 14 lie within 3.0 %: all ten pairs of periodic variants at C/10 lie within 3.5 %, while synthetic variants at C/10 differ by up to 43.7 %.
5. One dynamic protocol falls more than 8.7 % below a constant-current baseline: Periodic C at C/2, measured at 0.6 per hour, 10.2 % (5.4 to 14.7 %) below the constant current with a rest at 5 % SOC.
6. Scorecard: P1 held narrowly (81.3 % of gains above 8.7 % against more than 80 % predicted; 78.6 % with the authors' released lifetimes), P2 held (108.6 %, Synthetic 2a at C/2 against a rest at 100 % SOC), P3 and P4 failed; the paper's notebook processing changes no verdict, and an independent recomputation agrees. A comparison within the floor marks what duplicates can resolve at 90 % SOH, not an error in the paper. Archived as doi.org/10.5281/zenodo.23411591.